Mini Project 13 — Environmental Clearance Pre-Screening Service

Course: UE24AM342AA5 — ML System Design and AgentOps | **Domain:** Environment and Infrastructure Development

Component: Jackfruit (Level 3), 10 marks | **Team size:** 2–3 students

Technical terms used here are defined in `00_GLOSSARY.pdf`.

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1. Context

Before a project can be built in India it may need prior environmental clearance under the Environmental Impact Assessment (EIA) Notification. **Whether it needs clearance, and from whom, is decided before any assessment begins** — at a screening stage.

The screening answers two questions. Is the project in the Schedule at all. And is it **Category A**, appraised by the central ministry, or **Category B**, appraised by a state authority. The difference is months of timeline and a different depth of study.

Getting the category wrong is costly. A clearance granted at the wrong level can be set aside, sometimes after work has begun.

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2. What to Build

A multi-agent service that accepts a project description, validates it, determines Schedule applicability and category from the notification, and routes it with a cited justification.

This is a production service, not a notebook.

The system never determines clearance requirements authoritatively. It produces a screening opinion with supporting evidence. An accredited consultant decides.

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3. Corpus

The Ministry of Environment, Forest and Climate Change publishes the notification, every amendment and all office memoranda free on its website and on the PARIVESH portal. All documents below are free to download.

- The EIA Notification, 2006, including its Schedule and General Condition
- The amendments and office memoranda to that notification, from the PARIVESH portal

Prefer documents with a text layer. A scanned document needs optical character recognition (OCR) before it can be indexed.

Why this corpus is difficult. The Schedule is a table of thresholds, and the table is not the whole rule. A note printed after it, the General Condition, changes the category of some projects according to their location. It is not repeated in any Schedule row.

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4. What the System Must Do

#	Requirement
F1	Accept a project description over a documented REST interface, against a published **data contract**
F2	Enforce schema, range, categorical and cross-field validation at the **ingestion boundary**. Quarantine rejects.
F3	Classify the project against the Schedule sector and extract capacity, area, location and proximity entities
F4	Retrieve the applicable Schedule row **and the General Condition**
F5	Produce a **category determination grounded in retrieved provisions**, citing the Schedule entry and any condition applied

#	Requirement
F6	Route to one of: no-clearance-required, Category B to state authority, or Category A to central ministry. State the basis.
F7	**Escalate to a human instead of answering** when proximity is uncertain, amendments conflict, or the sector falls outside your corpus
F8	Expose health, metrics and structured logs for the whole pipeline

F5 and F7 hold most of the difficulty. A system that always answers is easy to build; one that knows when not to answer is not.

4.1 The difficult case

The system will be tested against this scenario at the demonstration.

"An industrial project that the Schedule places in Category B, located 2 kilometres from the boundary of a national park."

The Schedule row gives Category B, and that is what a search on the project type returns. But the General Condition, printed after the Schedule, treats Category B projects this close to a protected area as Category A. The system must retrieve the General Condition along with the Schedule row and cite both. **Reporting the Schedule category alone is the failure.**

At Evaluation 1, show the passages in your corpus that this case depends on, and the values you will test with. Once confirmed, this is the reference answer for your demonstration.

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5. Design Approach

The straightforward implementation embeds the request, retrieves the highest-scoring passages, and asks the model to answer from them. This handles ordinary cases but not the case in section 4.1.

The reason is structural. A single retrieval pass returns passages resembling the question asked. The case in section 4.1 turns on a provision the request never mentions, so a single embedding-based search is unlikely to return it.

The required approach is an agent that:

1. Retrieves the provision that appears to apply
2. Determines whether any other provision modifies it, and searches for such provisions
3. Searches again, using a query derived from what it has read
4. Verifies that the retrieved passages support the answer
5. Retries with a different strategy, or escalates, if they do not

Steps 2 and 4 distinguish an agent from a fixed pipeline. Both are assessed.

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6. Agent System Requirements

Unit 4 — agent architecture, orchestration, containerisation, observability.

#	Requirement
A1	Build the reasoning layer with **LangGraph or CrewAI**. Retrieval must be a **tool the agent chooses to call**, not a fixed stage. Provide at least three tools, such as corpus search, lookup of a clause by identifier, and lookup of the request's own fields. **Bound the loop** by steps, tool calls and wall-clock time.
A2	**At least two agents with distinct responsibilities** and a defined hand-off. Document the topology and what happens if one fails mid-run. Two prompts in one process are not two agents.
A3	**Verify every answer against the passages retrieved, before returning it.** Every citation must resolve to retrieved text. If verification fails, search again with a different strategy or escalate. An unverified answer must not be returned. This is the basis of F7.

#	Requirement
A4	Expose the service with **FastAPI**. Runs take tens of seconds, so **execute them asynchronously**: return a run identifier immediately, and the caller polls a status endpoint or receives a callback. Justify the choice. Place a queue or task runner between submission and execution.
A5	**Timeouts** on every model and tool call, **bounded retry with exponential backoff**, a **circuit breaker** on the provider, and **graceful degradation** along a defined fallback path. All must be demonstrable on request. A run that did not complete must not be returned as a confident answer.
A6	Track per agent and display on the dashboard: **task success rate**, steps and tool calls per run, latency end to end and per step, **retry rate**, container uptime.
A7	`docker compose up --scale <agent-worker>=3` must work. Measure the effect on throughput and p95, using the stub described in D6. If scaling does not help, identify the actual bottleneck.
A8	A simple **web client** that submits a request, obtains the result, and displays the recommendation with citations and the escalation path. No marks for appearance.

7. LLMOps Requirements

Unit 3 — retrieval, prompt management, evaluation, guardrails, monitoring.

#	Requirement
L1	Orchestrate retrieval with **LangChain or LlamaIndex**; index with **FAISS or Chroma**. Document the chunking and embedding strategy and how it addresses the difficulty in section 3. Re-embedding must be possible without rebuilding the whole index.
L2	Treat prompts as **versioned artifacts**, and record the prompt version with every response. Compare **at least two prompt variants** on the same evaluation set; report which performed better, on which metric, and by how much.
L3	Build a fixed evaluation set of **at least 20 scenarios**, each with its expected outcome and the passage that supports it, including the case in section 4.1, runnable as a batch from the command line. Use **TruLens** or equivalent. Report **groundedness, context relevance, faithfulness and hallucination rate**, stating how each is computed. Then make one deliberate change and **report all four metrics before and after**.
L4	**Prompt injection defence** on every free-text field reaching a model, **output filtering** so the system cannot issue a determination it is not authorised to make, and **toxicity detection** on input and output. Agents are more exposed than pipelines, because injected text can reach a model holding tools.
L5	Record **prompt tokens, completion tokens and cost at the time of the call**, attributed **per step as well as per run**. Report cost per 100 runs and the largest single cost driver.
L6	A **feedback endpoint** allowing the caller to mark a result satisfactory or not, with a reason, persisted and aggregated. **Score a small sample of live runs automatically** against the evaluation metrics, charting quality over time.

8. Platform and Deployment Requirements

Units 1 and 2 — data contracts, architecture, containerisation, observability, reliability.

#	Requirement
D1	Publish a **data contract** for every external interface. Enforce schema, type, range, category and cross-field validation at the **ingestion boundary**, before any agent starts. **Quarantine rejected records with their field-level errors.** A malformed request must not start an agent run.
D2	An **idempotent** pipeline that reads, parses, chunks, embeds and indexes the corpus, invocable from a command-line tool. Record **data freshness** — when the index was built, from which versions, how many chunks — and the pipeline's throughput, and expose both as metrics.
D3	**At least four services** under Docker Compose, starting with a **single** `docker compose up` from a clean clone, with dependencies and health checks declared. Nothing runs outside the containers. **Justify the placement of deployment boundaries** on scaling, failure isolation, independent release or resource profile.
D4	**No credentials, endpoints or model identifiers anywhere in the repository.** All environment-specific configuration injected at runtime, with a documented `.env.example`.

#	Requirement
D5	**Structured logging** throughout, a **correlation ID** across every service and agent step, tracing with **OpenTelemetry** exported to Jaeger or Grafana Tempo, **Prometheus** scraping every service, and a **Grafana** dashboard showing p50/p95 latency, error rate, throughput, availability and resource utilisation per container. **Given a run identifier, the entire run must be reconstructible** — every tool call, query, passage, prompt version, per-step latency and cost, and the verification outcome.
D6	**Declare a p95 service level objective (SLO) for a complete run before measuring anything**, justified against section 1. Measure a run from submission to final result, not the submission call alone. Load test with **Locust**, with the LLM replaced by a **stub service** that returns fixed responses after a realistic delay, so that the test measures your system rather than the provider's rate limits. Report RPS, p50 and p95 against the target and state whether it was met. Identify **the concurrency at which the target fails and the component responsible**. A target set after seeing the results does not satisfy this requirement.
D7	**Caching** — of embeddings, retrievals or whole runs, justified — and **rate limiting** per client. Report the measured effect of caching on p95 and cost per run. Rate limiting is a cost control here: one client looping against a multi-step agent can exhaust the budget in minutes.
D8	A **runbook** for the two most likely failure modes — symptom, confirmation, recovery — and an **incident log** with at least one genuine entry recording a real failure during development.

9. Demonstration

Prepare all ten items.

#	Item
1	`docker compose up` from a clean clone to a working system
2	A valid request producing a routed recommendation with citations that resolve to retrieved text
3	One run traced end to end — every tool call, query, passage, prompt version, per-step latency and cost
4	A malformed request rejected and quarantined, with evidence that no agent run started
5	A case the corpus cannot answer producing an escalation rather than a fabricated response
6	Self-verification catching an unsupported answer, followed by a further search or escalation
7	A prompt injection attempt and the guardrail response
8	The circuit breaker opening when the provider is unavailable, and the degradation path
9	Locust results against the declared target, the concurrency at which it fails, and the effect of `--scale`
10	The case in section 4.1, handled correctly

10. Assessment

If the system does not start and serve a valid request at the demonstration, the project is awarded zero marks, irrespective of the report, presentation or repository.

Establish a working end-to-end path by week two and add requirements after that. Completing all 22 requirements is not expected; sound prioritisation is itself assessed.

Criterion	Marks
Problem understanding, architecture and objective clarity	10
Presentation, teamwork and communication	10
Engineering judgement — tool selection, scope, handling of constraints	10
Report, code quality and repository documentation	10
Total (Evaluation 2, scaled to 8 marks)	**40**

11. Deliverables

Deliverable	Due
Corpus confirmation and project plan	Evaluation 1
Report and final GitHub repository	**One week before the presentation**
Live demonstration and viva	Evaluation 2

Report — a plain report covering, briefly: problem and scope; agent design (topology, tools, verification, loop limits); an architecture diagram with deployment boundaries marked; design decisions and what was rejected; evaluation results before and after the change; the p95 target, measured results, bottleneck and cost per 100 runs; and what did not work.

Repository — the README must be sufficient for someone unfamiliar with the project to run the system without assistance. This will be tested. Include the runbook, incident log, `.env.example` and evaluation set.

12. Cost

All software and data required are free. The corpus documents are public; **sentence-transformers run locally for embeddings at no cost**, and a small model such as all-MiniLM-L6-v2 keeps memory use low; FAISS, Chroma, LangGraph, CrewAI, LangChain, LlamaIndex, TruLens, FastAPI, Redis, Prometheus, Grafana, OpenTelemetry, Jaeger and Locust are all free. For the model, use a provider's free tier. A local model via Ollama is also possible but needs a considerably more powerful machine.

Embed locally. Embedding a corpus through a paid API is the only operation here with a real cost.

Agents multiply model calls — typically several per run, against one for a fixed pipeline. Free tiers also have low rate limits, which is why load testing uses a stub (D6). This is the reason for the loop limits in A1, the caching in D7 and the per-step cost tracking in L5. Set a spending limit on the account before starting, and place the model behind an interface you control so the provider can be changed by configuration.

13. Out of Scope

- Model training is not required.
- Cloud deployment is not required; the system runs locally under Docker Compose.
- **Do not attempt impact assessment.** Screening and categorisation only.